```
# ====================================================================
# Introduction to Data Science - Wine Dataset Analysis
# Authors: Hemanth Jadiswami Prabhakaran (7026000) & Manoj Kumar
Prabhakaran (7026006)
# Assignment: Analysis of Portuguese Wine Dataset
# Date: 30th June 2025
# ====================================================================

# Clear workspace
rm(list = ls())

install.packages(c("reshape2", "moments", "car", "pROC", "psych"))
# Load necessary libraries

Warning message:
"packages 'reshape2', 'moments' are in use and will not be installed"

also installing the dependencies 'rbibutils', 'cowplot', 'Deriv',
'microbenchmark', 'Rdpack', 'numDeriv', 'doBy', 'SparseM',
'MatrixModels', 'minqa', 'nloptr', 'reformulas', 'RcppEigen',
'carData', 'abind', 'Formula', 'pbkrtest', 'quantreg', 'lme4'



package 'rbibutils' successfully unpacked and MD5 sums checked
package 'cowplot' successfully unpacked and MD5 sums checked
package 'Deriv' successfully unpacked and MD5 sums checked
package 'microbenchmark' successfully unpacked and MD5 sums checked
package 'Rdpack' successfully unpacked and MD5 sums checked
package 'numDeriv' successfully unpacked and MD5 sums checked
package 'doBy' successfully unpacked and MD5 sums checked
package 'SparseM' successfully unpacked and MD5 sums checked
package 'MatrixModels' successfully unpacked and MD5 sums checked
package 'minqa' successfully unpacked and MD5 sums checked
package 'nloptr' successfully unpacked and MD5 sums checked
package 'reformulas' successfully unpacked and MD5 sums checked
package 'RcppEigen' successfully unpacked and MD5 sums checked
package 'carData' successfully unpacked and MD5 sums checked
package 'abind' successfully unpacked and MD5 sums checked
package 'Formula' successfully unpacked and MD5 sums checked
package 'pbkrtest' successfully unpacked and MD5 sums checked
package 'quantreg' successfully unpacked and MD5 sums checked
package 'lme4' successfully unpacked and MD5 sums checked
package 'car' successfully unpacked and MD5 sums checked
package 'pROC' successfully unpacked and MD5 sums checked
package 'psych' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
	C:\Users\Manoj Kumar\AppData\Local\Temp\Rtmp69OsAb\
downloaded_packages
```

```r
# Load required packages
# Note: Install packages if not already installed using
install.packages()
library(reshape2)  # For data manipulation (taught in class)
library(moments)   # For skewness calculation
library(car)       # For regression diagnostics
library(pROC)      # For ROC curves and AUC
library(psych)     # For factor analysis and descriptive statistics
```

```
Loading required package: carData

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

    cov, smooth, var

Attaching package: 'psych'

The following object is masked from 'package:car':

    logit
```

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# TASK 1: DESCRIPTIVE STATISTICS AND DATA EXPLORATION

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```
# Read the wine dataset
wine_data <- read.csv("D:\\DataScience\\wine.csv")

# Basic data structure
cat("Dataset Structure:\n")
str(wine_data)
cat("\nDataset Dimensions:", dim(wine_data), "\n")

Dataset Structure:
'data.frame':   6497 obs. of  14 variables:
 $ X                   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ fixed.acidity       : num  7.4 7.8 7.8 11.2 7.4 7.4 7.9 7.3 7.8 7.5
...
 $ volatile.acidity    : num  0.7 0.88 0.76 0.28 0.7 0.66 0.6 0.65
0.58 0.5 ...
 $ citric.acid         : num  0 0 0.04 0.56 0 0 0.06 0 0.02 0.36 ...
 $ residual.sugar      : num  1.9 2.6 2.3 1.9 1.9 1.8 1.6 1.2 2
6.1 ...
 $ chlorides           : num  0.076 0.098 0.092 0.075 0.076 0.075
0.069 0.065 0.073 0.071 ...
 $ free.sulfur.dioxide : num  11 25 15 17 11 13 15 15 9 17 ...
 $ total.sulfur.dioxide: num  34 67 54 60 34 40 59 21 18 102 ...
 $ density             : num  0.998 0.997 0.997 0.998 0.998 ...
 $ pH                  : num  3.51 3.2 3.26 3.16 3.51 3.51 3.3 3.39
3.36 3.35 ...
 $ sulphates           : num  0.56 0.68 0.65 0.58 0.56 0.56 0.46 0.47
0.57 0.8 ...
 $ alcohol             : num  9.4 9.8 9.8 9.8 9.4 9.4 9.4 10 9.5
10.5 ...
 $ quality             : int  5 5 5 6 5 5 5 7 7 5 ...
 $ variety             : chr  "red" "red" "red" "red" ...

Dataset Dimensions: 6497 14
```

```r
# 1a) Distribution parameters for all metric variables
cat("\n=== TASK 1A: DESCRIPTIVE STATISTICS ===\n")

# Identify metric (numeric) and categorical variables
metric_vars <- c("fixed.acidity", "volatile.acidity", "citric.acid",
                 "residual.sugar", "chlorides", "free.sulfur.dioxide",
                 "total.sulfur.dioxide", "density", "pH", "sulphates",

                 "alcohol", "quality")

categorical_vars <- c("variety")
```

```
=== TASK 1A: DESCRIPTIVE STATISTICS ===
```

```r
# Create comprehensive descriptive statistics table
desc_stats <- data.frame(
  Variable = character(),
  Mean = numeric(),
  SD = numeric(),
  Min = numeric(),
  Q1 = numeric(),
  Median = numeric(),
  Q3 = numeric(),
  Max = numeric(),
  Missing = numeric(),
  Skewness = numeric(),
  stringsAsFactors = FALSE
)

# Calculate statistics for each metric variable
for(var in metric_vars) {
  if(var %in% names(wine_data)) {
    x <- wine_data[[var]]
    desc_stats <- rbind(desc_stats, data.frame(
      Variable = var,
      Mean = round(mean(x, na.rm = TRUE), 3),
      SD = round(sd(x, na.rm = TRUE), 3),
      Min = round(min(x, na.rm = TRUE), 3),
      Q1 = round(quantile(x, 0.25, na.rm = TRUE), 3),
      Median = round(median(x, na.rm = TRUE), 3),
      Q3 = round(quantile(x, 0.75, na.rm = TRUE), 3),
      Max = round(max(x, na.rm = TRUE), 3),
      Missing = sum(is.na(x)),
      Skewness = round(skewness(x, na.rm = TRUE), 3)
    ))
  }
}

print(desc_stats)
```

```r
# Frequency distributions for categorical variables
cat("\n=== FREQUENCY DISTRIBUTIONS FOR CATEGORICAL VARIABLES ===\n")
for(var in categorical_vars) {
  if(var %in% names(wine_data)) {
    cat("\n", var, ":\n")
    freq_table <- table(wine_data[[var]], useNA = "ifany")
    print(freq_table)
    print(prop.table(freq_table))
  }
}
```

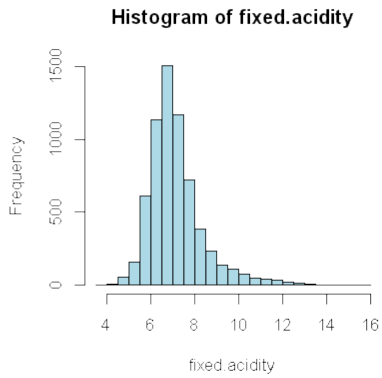
```
                   Variable    Mean     SD   Min     Q1  Median      Q3
Max
25%          fixed.acidity   7.215  1.296 3.800  6.400   7.000   7.700
15.900
25%1      volatile.acidity   0.340  0.165 0.080  0.230   0.290   0.400
1.580
25%2           citric.acid   0.319  0.145 0.000  0.250   0.310   0.390
1.660
25%3        residual.sugar   5.443  4.758 0.600  1.800   3.000   8.100
65.800
25%4             chlorides   0.056  0.035 0.009  0.038   0.047   0.065
0.611
25%5   free.sulfur.dioxide  30.525 17.749 1.000 17.000  29.000  41.000
289.000
25%6  total.sulfur.dioxide 115.745 56.522 6.000 77.000 118.000 156.000
440.000
25%7               density   0.995  0.003 0.987  0.992   0.995   0.997
1.039
25%8                    pH   3.219  0.161 2.720  3.110   3.210   3.320
4.010
25%9             sulphates   0.531  0.149 0.220  0.430   0.510   0.600
2.000
25%10              alcohol  10.492  1.193 8.000  9.500  10.300  11.300
14.900
25%11              quality   5.818  0.873 3.000  5.000   6.000   6.000
9.000
      Missing Skewness
25%         0    1.723
25%1        0    1.495
25%2        0    0.472
25%3        0    1.435
25%4        0    5.399
25%5        0    1.220
25%6        0   -0.001
25%7        0    0.503
25%8        0    0.387
25%9        0    1.797
25%10       0    0.566
25%11       0    0.190
```

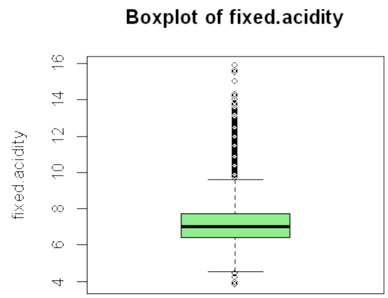
```
=== FREQUENCY DISTRIBUTIONS FOR CATEGORICAL VARIABLES ===

 variety :

  red white
 1599  4898

      red     white
0.2461136 0.7538864
```

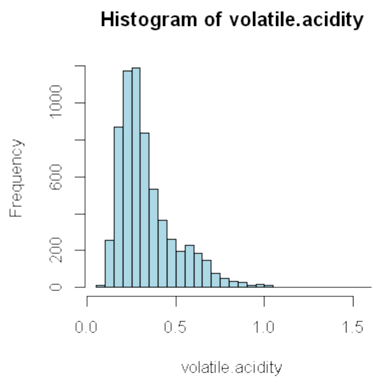
```r
# 1b) Create suitable graphics for all variables
cat("\n=== TASK 1B: GRAPHICS AND DISTRIBUTION ASSESSMENT ===\n")
```

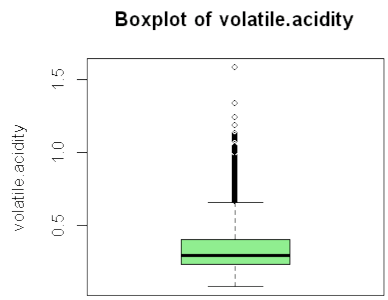
```
=== TASK 1B: GRAPHICS AND DISTRIBUTION ASSESSMENT ===
```

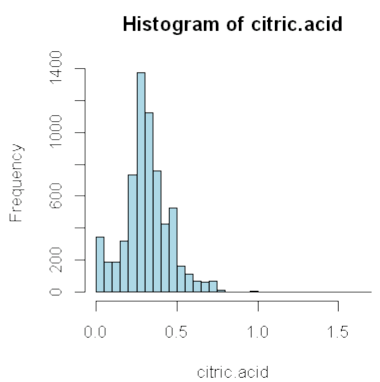
```r
# Set up graphics parameters
par(mfrow = c(2, 2))

# Create histograms and boxplots for metric variables
for(var in metric_vars) {
  if(var %in% names(wine_data)) {
    # Histogram
    hist(wine_data[[var]], main = paste("Histogram of", var),
         xlab = var, col = "lightblue", breaks = 30)

    # Boxplot
    boxplot(wine_data[[var]], main = paste("Boxplot of", var),
            ylab = var, col = "lightgreen")
  }
}
```

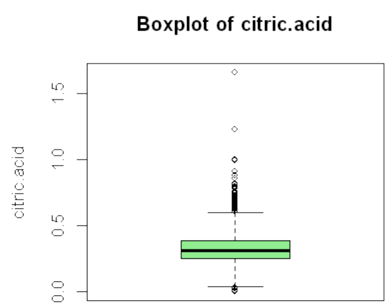

Histogram of fixed.acidity
Frequency
fixed.acidity

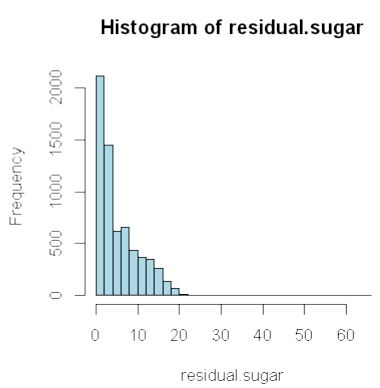

Boxplot of fixed.acidity
fixed.acidity

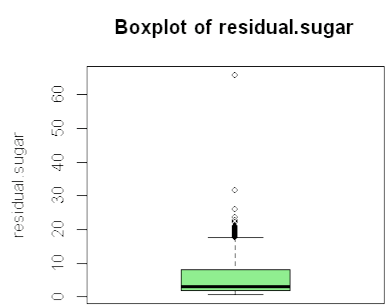

Histogram of volatile.acidity
Frequency
volatile.acidity

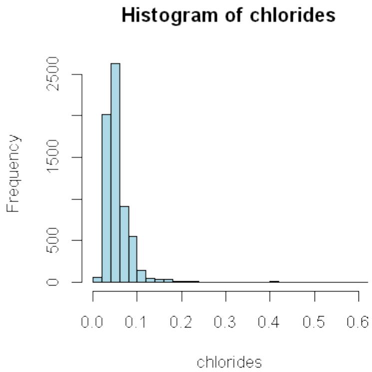

Boxplot of volatile.acidity
volatile.acidity

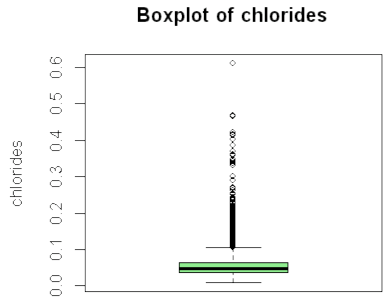

Histogram of citric.acid
Frequency
citric.acid

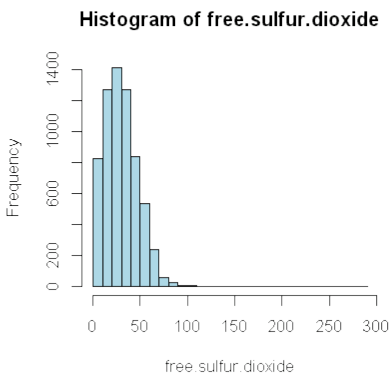

Boxplot of citric.acid
citric.acid

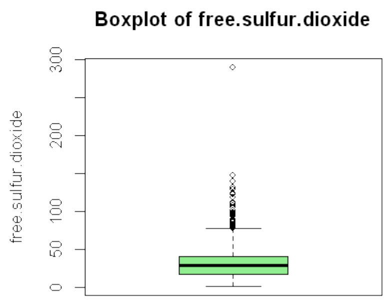

Histogram of residual.sugar
Frequency
residual.sugar

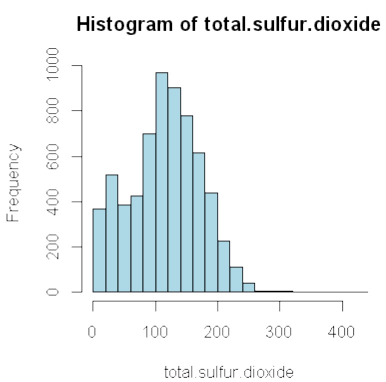

Boxplot of residual.sugar
residual.sugar

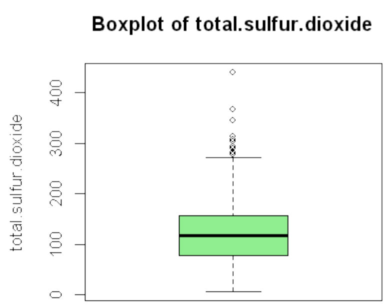

Histogram of chlorides
Frequency
0
500
1500
2500
0.0
0.1
0.2
0.3
0.4
0.5
0.6
chlorides

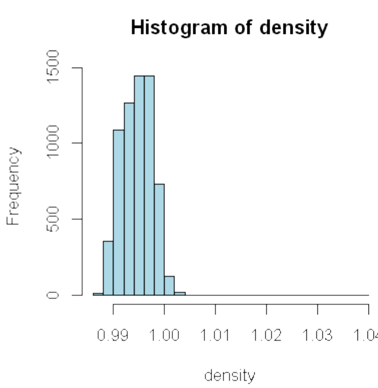

Boxplot of chlorides
chlorides
0.0
0.1
0.2
0.3
0.4
0.5
0.6

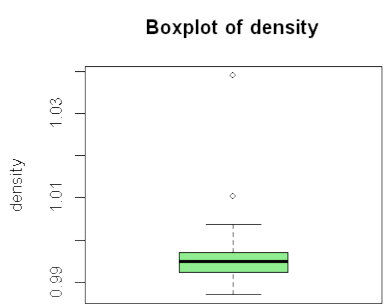

Histogram of free.sulfur.dioxide
Frequency
0
200
600
1000
1400
0
50
100
150
200
250
300
free.sulfur.dioxide

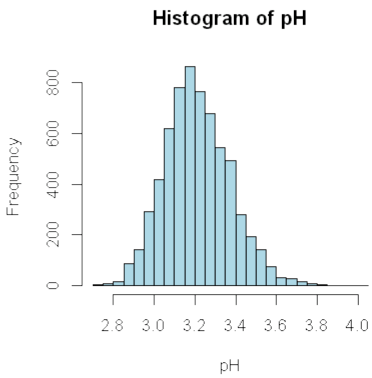

Boxplot of free.sulfur.dioxide
free.sulfur.dioxide
0
50
100
200
300

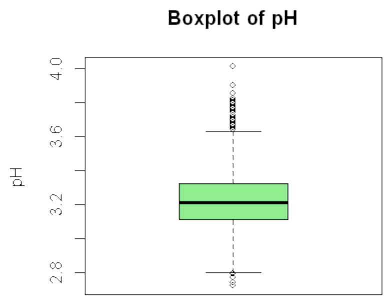

Histogram of total.sulfur.dioxide
Frequency
0
200
400
600
800
1000
0
100
200
300
400
total.sulfur.dioxide

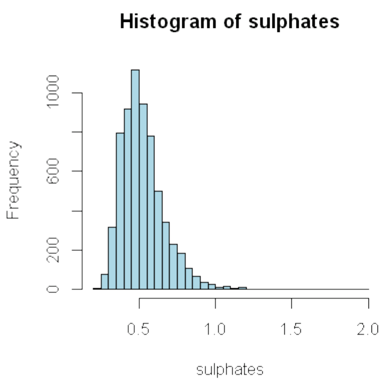

Boxplot of total.sulfur.dioxide
total.sulfur.dioxide
0
100
200
300
400

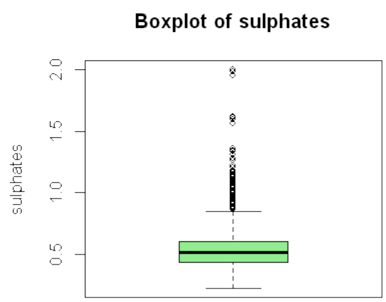

Histogram of density
Frequency
0
500
1000
1500
0.99
1.00
1.01
1.02
1.03
1.04
density

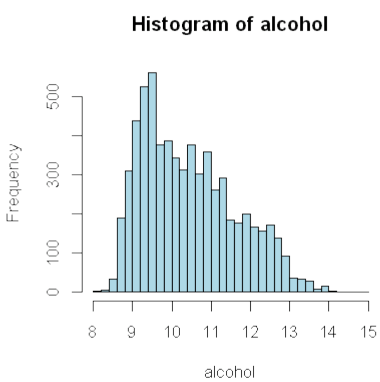

Boxplot of density
density
0.99
1.01
1.03

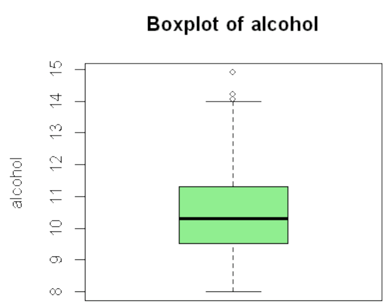

Histogram of pH
Frequency
0
200
400
600
800
2.8
3.0
3.2
3.4
3.6
3.8
4.0
pH

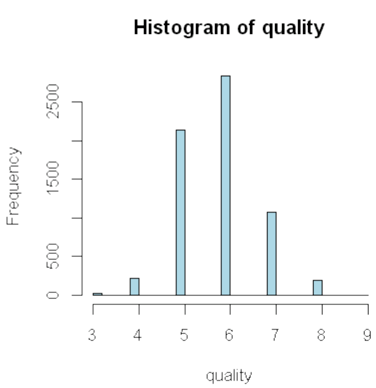

Boxplot of pH
pH
2.8
3.2
3.6
4.0

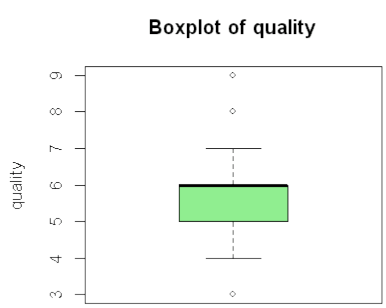

Histogram of sulphates
Frequency
0
200
600
1000
0.5
1.0
1.5
2.0
sulphates


Boxplot of sulphates
sulphates
0.5
1.0
1.5
2.0

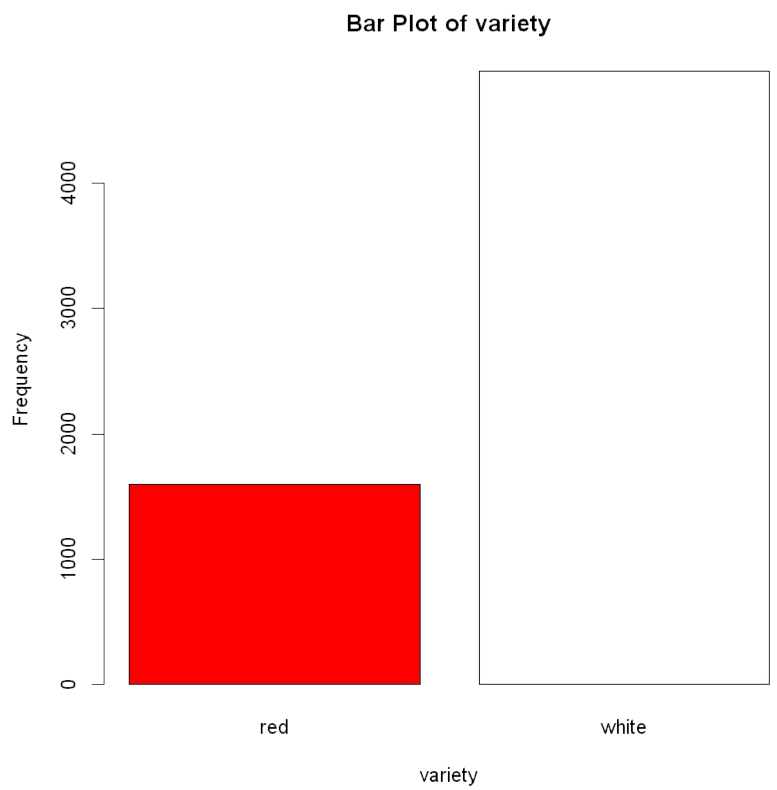

Histogram of alcohol
Frequency
0
100
300
500
8
9
10
11
12
13
14
15
alcohol


Boxplot of alcohol
alcohol
8
9
10
11
12
13
14
15


Histogram of quality
Frequency
0
500
1500
2500
3
4
5
6
7
8
9
quality


Boxplot of quality
quality
3
4
5
6
7
8
9

```r
# Bar plot for categorical variables
for(var in categorical_vars) {
  if(var %in% names(wine_data)) {
    barplot(table(wine_data[[var]]), main = paste("Bar Plot of", var),
            xlab = var, ylab = "Frequency", col = c("red", "white"))
  }
}
```




```r
# Reset graphics parameters
par(mfrow = c(1, 1))
```

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# TASK 2: T-TEST FOR ALCOHOL CONTENT BETWEEN RED AND WHITE WINES

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```
cat("\n=== TASK 2: T-TEST ANALYSIS ===\n")
```

```
=== TASK 2: T-TEST ANALYSIS ===
```

```
# Separate alcohol content by wine variety
red_alcohol <- wine_data$alcohol[wine_data$variety == "red"]
white_alcohol <- wine_data$alcohol[wine_data$variety == "white"]

# Check t-test assumptions
cat("T-Test Assumption Checks:\n")
```

```
T-Test Assumption Checks:
```

```
# 1. Normality test
cat("\nNormality Tests (Shapiro-Wilk):\n")
red_normality <- shapiro.test(sample(red_alcohol, min(5000,
length(red_alcohol))))
white_normality <- shapiro.test(sample(white_alcohol, min(5000,
length(white_alcohol))))

cat("Red wines alcohol normality p-value:", red_normality$p.value, "\
n")
cat("White wines alcohol normality p-value:", white_normality$p.value,
"\n")
```

```
Normality Tests (Shapiro-Wilk):
Red wines alcohol normality p-value: 6.644057e-27
White wines alcohol normality p-value: 2.569014e-36
```

```
# 2. Equal variances test
var_test <- var.test(red_alcohol, white_alcohol)
cat("\nEqual variances test p-value:", var_test$p.value, "\n")
```

```
Equal variances test p-value: 5.947444e-12

# Perform appropriate t-test
if(var_test$p.value < 0.05) {
  # Unequal variances
  t_result <- t.test(red_alcohol, white_alcohol, var.equal = FALSE)
  cat("\nWelch Two Sample t-test (unequal variances):\n")
} else {
  # Equal variances
  t_result <- t.test(red_alcohol, white_alcohol, var.equal = TRUE)
  cat("\nTwo Sample t-test (equal variances):\n")
}

print(t_result)


Welch Two Sample t-test (unequal variances):

     Welch Two Sample t-test

data:  red_alcohol and white_alcohol
t = -2.859, df = 3100.5, p-value = 0.004278
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.15388669 -0.02868117
sample estimates:
mean of x mean of y
 10.42298  10.51427


# Effect size (Cohen's d)
pooled_sd <- sqrt(((length(red_alcohol)-1)*var(red_alcohol) +
                   (length(white_alcohol)-1)*var(white_alcohol)) /
                  (length(red_alcohol) + length(white_alcohol) - 2))
cohens_d <- (mean(red_alcohol) - mean(white_alcohol)) / pooled_sd
cat("Cohen's d (effect size):", cohens_d, "\n")

Cohen's d (effect size): -0.07657052
```

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## TASK 3: LINEAR REGRESSION FOR RED WINES QUALITY

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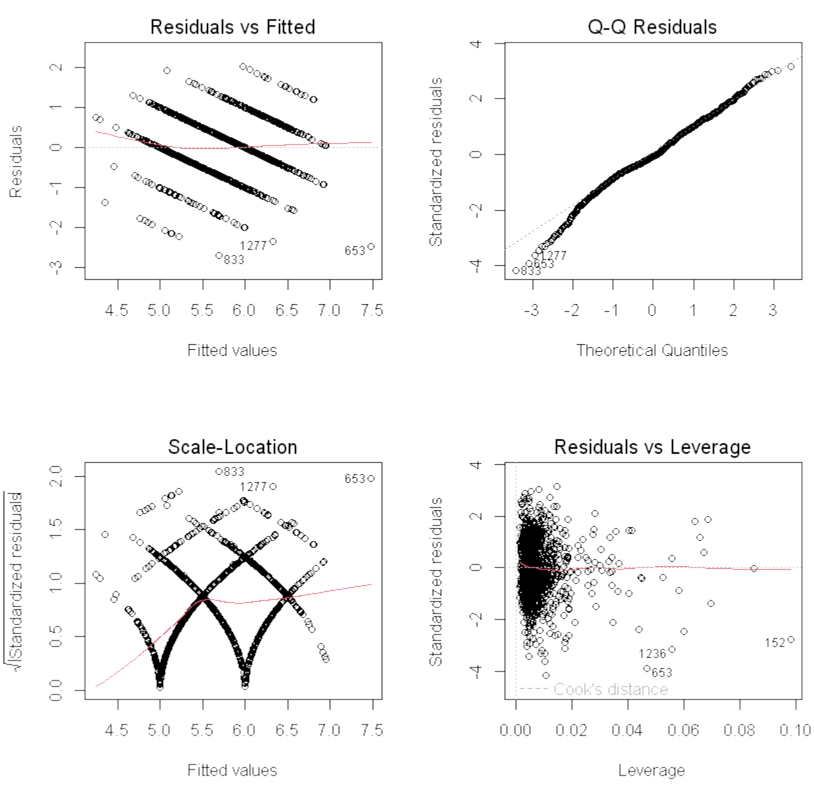
```
cat("\n=== TASK 3: LINEAR REGRESSION ANALYSIS (RED WINES ONLY) ===\n")

# Filter for red wines only
red_wines <- wine_data[wine_data$variety == "red", ]

# Remove non-predictor variables
predictor_vars <- c("fixed.acidity", "volatile.acidity",
"citric.acid",
                    "residual.sugar", "chlorides",
"free.sulfur.dioxide",
                    "total.sulfur.dioxide", "density", "pH",
"sulphates", "alcohol")

# Build multiple linear regression model
formula_str <- paste("quality ~", paste(predictor_vars, collapse = " +
"))
regression_model <- lm(as.formula(formula_str), data = red_wines)

# Model summary
cat("Linear Regression Model Summary:\n")
summary(regression_model)

# Regression diagnostics
cat("\n=== REGRESSION DIAGNOSTICS ===\n")

# Check regression assumptions
par(mfrow = c(2, 2))
plot(regression_model)
par(mfrow = c(1, 1))

# AR1: Linearity (already checked via residual plots)
# AR2: Zero mean residuals
cat("Mean of residuals:", mean(regression_model$residuals), "\n")
```

```r
# AR3: No autocorrelation (Durbin-Watson test)
dw_test <- car::durbinWatsonTest(regression_model)
cat("Durbin-Watson test p-value:", dw_test$p, "\n")

# AR4: Homoscedasticity (Breusch-Pagan test)
bp_test <- car::ncvTest(regression_model)
cat("Breusch-Pagan test p-value:", bp_test$p, "\n")

# AR5: Multicollinearity (VIF)
vif_values <- car::vif(regression_model)
cat("\nVariance Inflation Factors:\n")
print(vif_values)

# AR6: Normality of residuals
shapiro_residuals <- shapiro.test(sample(regression_model$residuals,
                                        min(5000,
length(regression_model$residuals))))
cat("Normality of residuals p-value:", shapiro_residuals$p.value, "\
n")
```

```
=== TASK 3: LINEAR REGRESSION ANALYSIS (RED WINES ONLY) ===
Linear Regression Model Summary:


Call:
lm(formula = as.formula(formula_str), data = red_wines)

Residuals:
     Min       1Q   Median       3Q      Max 
-2.68911 -0.36652 -0.04699  0.45202  2.02498 

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)           2.197e+01  2.119e+01   1.036   0.3002    
fixed.acidity         2.499e-02  2.595e-02   0.963   0.3357    
volatile.acidity     -1.084e+00  1.211e-01  -8.948  < 2e-16 ***
citric.acid          -1.826e-01  1.472e-01  -1.240   0.2150    
residual.sugar        1.633e-02  1.500e-02   1.089   0.2765    
chlorides            -1.874e+00  4.193e-01  -4.470 8.37e-06 ***
free.sulfur.dioxide   4.361e-03  2.171e-03   2.009   0.0447 *  
total.sulfur.dioxide -3.265e-03  7.287e-04  -4.480 8.00e-06 ***
density              -1.788e+01  2.163e+01  -0.827   0.4086    
pH                   -4.137e-01  1.916e-01  -2.159   0.0310 *  
sulphates             9.163e-01  1.143e-01   8.014 2.13e-15 ***
alcohol               2.762e-01  2.648e-02  10.429  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.648 on 1587 degrees of freedom
```

```
Multiple R-squared:  0.3606,	Adjusted R-squared:  0.3561 
F-statistic: 81.35 on 11 and 1587 DF,  p-value: < 2.2e-16


=== REGRESSION DIAGNOSTICS ===
Mean of residuals: -3.777011e-17 
Durbin-Watson test p-value: 0 
Breusch-Pagan test p-value: 2.041867e-06 

Variance Inflation Factors:
       fixed.acidity     volatile.acidity          citric.acid 
            7.767512             1.789390             3.128022 
      residual.sugar            chlorides  free.sulfur.dioxide 
            1.702588             1.481932             1.963019 
total.sulfur.dioxide              density                   pH 
            2.186813             6.343760             3.329732 
           sulphates              alcohol 
            1.429434             3.031160 
Normality of residuals p-value: 1.95424e-08 
```



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# TASK 4: CLASSIFICATION - GOOD VS BAD WINES

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```
cat("\n=== TASK 4: WINE QUALITY CLASSIFICATION ===\n")

# Create binary quality variable (good = quality >= 8, bad = quality
<= 4)
wine_data$quality_binary <- ifelse(wine_data$quality >= 8, "good",
                                  ifelse(wine_data$quality <= 4,
"bad", "medium"))

# Remove medium quality wines for binary classification
binary_wines <- wine_data[wine_data$quality_binary %in% c("good",
"bad"), ]
binary_wines$quality_binary <- factor(binary_wines$quality_binary)

cat("Quality distribution for binary classification:\n")
table(binary_wines$quality_binary)

# Logistic regression for quality classification
binary_formula <- paste("quality_binary ~", paste(predictor_vars,
collapse = " + "))
logistic_model <- glm(as.formula(binary_formula),
                     data = binary_wines, family = binomial())

cat("\nLogistic Regression Model Summary:\n")
summary(logistic_model)

# Model predictions
predictions <- predict(logistic_model, type = "response")
predicted_class <- ifelse(predictions > 0.5, "good", "bad")

# Confusion matrix
confusion_matrix <- table(Predicted = predicted_class,
                         Actual = binary_wines$quality_binary)
cat("\nConfusion Matrix:\n")
print(confusion_matrix)
```

```r
# Calculate accuracy, precision, recall
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)
precision <- confusion_matrix[2,2] / sum(confusion_matrix[2,])
recall <- confusion_matrix[2,2] / sum(confusion_matrix[,2])
f1_score <- 2 * (precision * recall) / (precision + recall)

cat("Accuracy:", round(accuracy, 3), "\n")
cat("Precision:", round(precision, 3), "\n")
cat("Recall:", round(recall, 3), "\n")
cat("F1-Score:", round(f1_score, 3), "\n")
```

```
=== TASK 4: WINE QUALITY CLASSIFICATION ===
Quality distribution for binary classification:


 bad good
 246  198


Logistic Regression Model Summary:


Call:
glm(formula = as.formula(binary_formula), family = binomial(),
    data = binary_wines)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)           3.609e+02  2.044e+02   1.766  0.07747 .
fixed.acidity         3.912e-01  2.447e-01   1.598  0.10994
volatile.acidity     -8.784e+00  1.551e+00  -5.662 1.49e-08 ***
citric.acid          -2.680e-01  1.411e+00  -0.190  0.84935
residual.sugar        3.779e-01  9.161e-02   4.125 3.71e-05 ***
chlorides            -3.125e-01  5.477e+00  -0.057  0.95450
free.sulfur.dioxide   2.896e-02  9.659e-03   2.998  0.00272 **
total.sulfur.dioxide -1.109e-02  4.660e-03  -2.379  0.01734 *
density              -3.879e+02  2.082e+02  -1.863  0.06242 .
pH                    2.918e+00  1.360e+00   2.145  0.03195 *
sulphates             3.339e+00  1.309e+00   2.550  0.01077 *
alcohol               1.101e+00  2.890e-01   3.808  0.00014 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 610.32  on 443  degrees of freedom
Residual deviance: 302.18  on 432  degrees of freedom
AIC: 326.18
```

```
Number of Fisher Scoring iterations: 6


Confusion Matrix:
         Actual
Predicted bad good
     bad  215   36
     good  31  162
Accuracy: 0.849
Precision: 0.839
Recall: 0.818
F1-Score: 0.829
```

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# TASK 5: COLOR PREDICTION WITH TRAIN/TEST SPLIT

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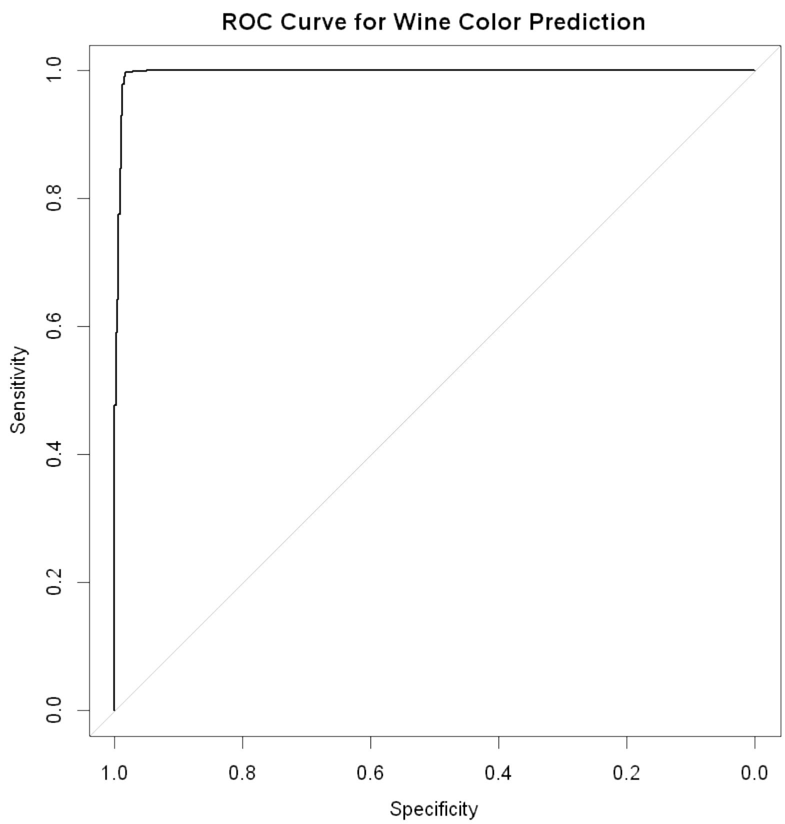
```r
cat("\n=== TASK 5: WINE COLOR PREDICTION ===\n")

# Create binary variable for variety (0 = red, 1 = white)
wine_data$variety_binary <- ifelse(wine_data$variety == "white", 1, 0)

# Train/test split (70/30)
set.seed(123)  # For reproducibility
train_indices <- sample(nrow(wine_data), 0.7 * nrow(wine_data))
train_data <- wine_data[train_indices, ]
test_data <- wine_data[-train_indices, ]

cat("Training set size:", nrow(train_data), "\n")
cat("Test set size:", nrow(test_data), "\n")

# Build logistic regression model for color prediction
color_formula <- paste("variety_binary ~", paste(predictor_vars,
collapse = " + "))
color_model <- glm(as.formula(color_formula),
```

```r
                   data = train_data, family = binomial())

cat("\nColor Prediction Model Summary:\n")
summary(color_model)

# Predictions on test set
test_predictions <- predict(color_model, newdata = test_data, type =
"response")
test_predicted_class <- ifelse(test_predictions > 0.5, 1, 0)

# Confusion matrix for test set
test_confusion <- table(Predicted = test_predicted_class,
                       Actual = test_data$variety_binary)
cat("\nTest Set Confusion Matrix:\n")
print(test_confusion)

# Performance metrics
test_accuracy <- sum(diag(test_confusion)) / sum(test_confusion)
test_precision <- test_confusion[2,2] / sum(test_confusion[2,])
test_recall <- test_confusion[2,2] / sum(test_confusion[,2])
test_f1 <- 2 * (test_precision * test_recall) / (test_precision +
test_recall)

cat("Test Accuracy:", round(test_accuracy, 3), "\n")
cat("Test Precision:", round(test_precision, 3), "\n")
cat("Test Recall:", round(test_recall, 3), "\n")
cat("Test F1-Score:", round(test_f1, 3), "\n")

# ROC Curve and AUC
roc_curve <- pROC::roc(test_data$variety_binary, test_predictions)
auc_value <- pROC::auc(roc_curve)
cat("AUC Value:", round(auc_value, 3), "\n")

# Plot ROC curve
plot(roc_curve, main = "ROC Curve for Wine Color Prediction")
```

```
=== TASK 5: WINE COLOR PREDICTION ===
Training set size: 4547
Test set size: 1950

Color Prediction Model Summary:


Call:
glm(formula = as.formula(color_formula), family = binomial(),
    data = train_data)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)           1.341e+03  1.796e+02   7.467 8.22e-14 ***
```

```
fixed.acidity        -2.829e-01  2.544e-01  -1.112 0.266094    
volatile.acidity     -7.296e+00  1.182e+00  -6.172 6.74e-10 ***
citric.acid           3.238e+00  1.365e+00   2.371 0.017719 *  
residual.sugar        8.842e-01  1.139e-01   7.764 8.23e-15 ***
chlorides            -2.453e+01  4.575e+00  -5.360 8.32e-08 ***
free.sulfur.dioxide  -6.345e-02  1.688e-02  -3.758 0.000171 ***
total.sulfur.dioxide  5.552e-02  5.921e-03   9.376  < 2e-16 ***
density              -1.330e+03  1.835e+02  -7.245 4.34e-13 ***
pH                   -7.346e-01  1.534e+00  -0.479 0.632040    
sulphates            -4.849e+00  1.440e+00  -3.366 0.000762 ***
alcohol              -1.250e+00  2.712e-01  -4.610 4.02e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 5065.53  on 4546  degrees of freedom
Residual deviance:  308.77  on 4535  degrees of freedom
AIC: 332.77

Number of Fisher Scoring iterations: 10


Test Set Confusion Matrix:
         Actual
Predicted    0    1
        0  471    2
        1   13 1464
Test Accuracy: 0.992
Test Precision: 0.991
Test Recall: 0.999
Test F1-Score: 0.995

Setting levels: control = 0, case = 1

Setting direction: controls < cases


AUC Value: 0.996
```




# ================================================================

# TASK 6: FACTOR ANALYSIS

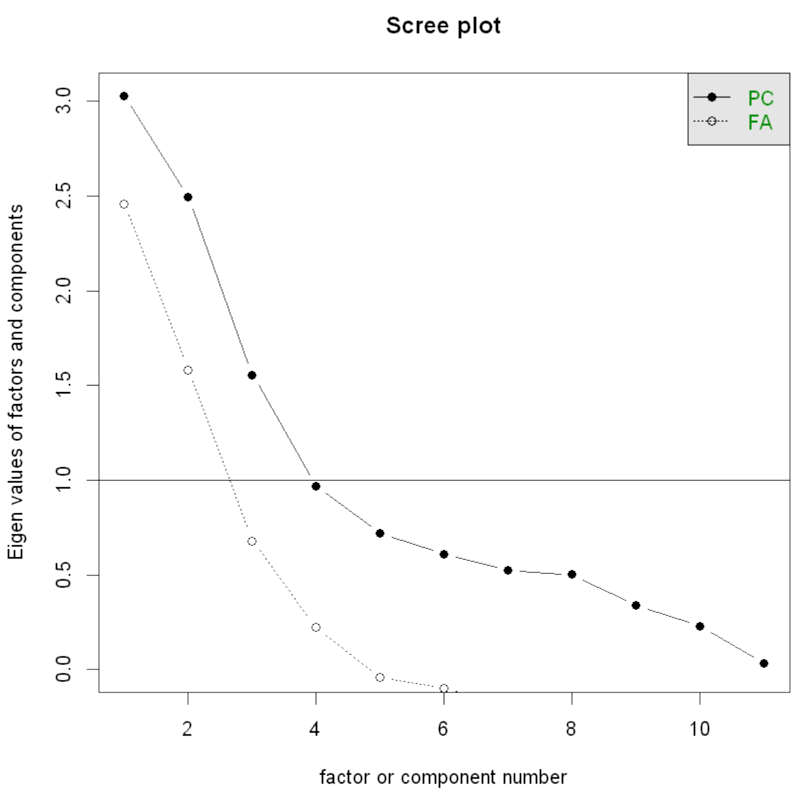
# ================================================================

```
cat("\n=== TASK 6: FACTOR ANALYSIS ===\n")

# Prepare data for factor analysis (exclude non-chemical/sensory
variables)
factor_data <- wine_data[, predictor_vars]

# Remove any rows with missing values
factor_data <- na.omit(factor_data)
```

```r
# Check correlation matrix
correlation_matrix <- cor(factor_data)
cat("Correlation Matrix (first 5x5):\n")
print(correlation_matrix[1:5, 1:5])

# Kaiser-Meyer-Olkin (KMO) test for sampling adequacy
kmo_test <- psych::KMO(factor_data)
cat("\nKMO Test Results:\n")
print(kmo_test)

# Bartlett's test of sphericity
bartlett_test <- psych::cortest.bartlett(correlation_matrix, n =
nrow(factor_data))
cat("\nBartlett's Test p-value:", bartlett_test$p.value, "\n")

# Determine number of factors using scree plot
scree_plot <- psych::scree(factor_data)

# Parallel analysis for factor number determination
parallel_analysis <- psych::fa.parallel(factor_data, fm = "ml", fa =
"fa")

# Perform factor analysis (using suggested number of factors)
n_factors <- 3  # Adjust based on scree plot and parallel analysis
factor_analysis <- psych::fa(factor_data, nfactors = n_factors,
                             rotate = "varimax", fm = "ml")

cat("\nFactor Analysis Results:\n")
print(factor_analysis)

# Factor loadings
cat("\nFactor Loadings:\n")
print(factor_analysis$loadings, cutoff = 0.3)
```

```
=== TASK 6: FACTOR ANALYSIS ===
Correlation Matrix (first 5x5):
                 fixed.acidity volatile.acidity citric.acid
residual.sugar
fixed.acidity        1.0000000        0.2190083  0.32443573     -
0.1119813
volatile.acidity     0.2190083        1.0000000 -0.37798132     -
0.1960112
citric.acid          0.3244357       -0.3779813  1.00000000
0.1424512
residual.sugar      -0.1119813       -0.1960112  0.14245123
1.0000000
chlorides            0.2981948        0.3771243  0.03899801     -
0.1289405
                   chlorides
```

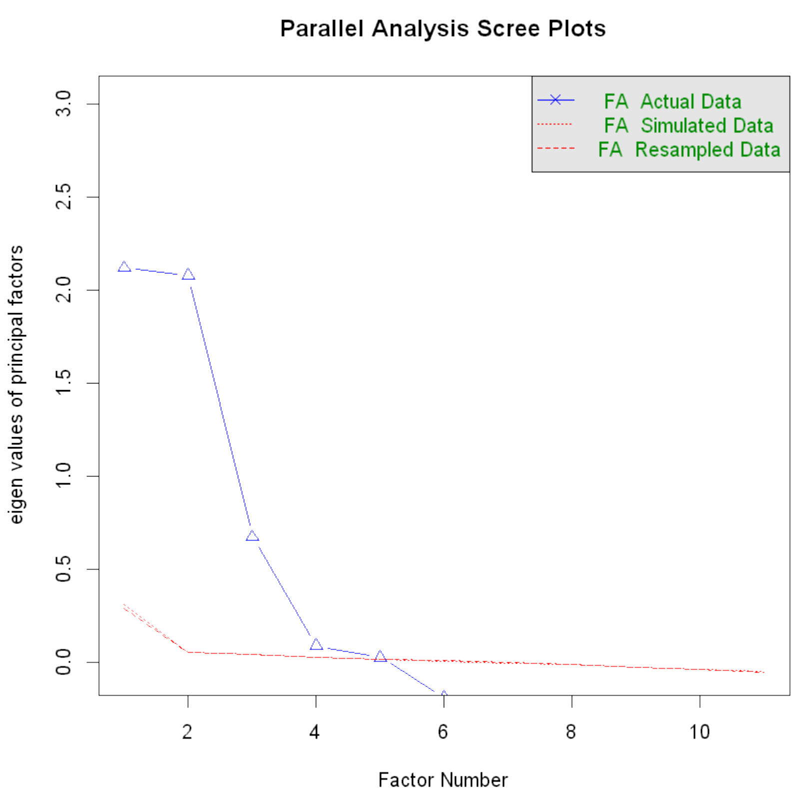
```
fixed.acidity     0.29819477
volatile.acidity  0.37712428
citric.acid       0.03899801
residual.sugar   -0.12894050
chlorides         1.00000000

KMO Test Results:
Kaiser-Meyer-Olkin factor adequacy
Call: psych::KMO(r = factor_data)
Overall MSA =  0.41
MSA for each item = 
       fixed.acidity     volatile.acidity          citric.acid 
                0.28                 0.61                 0.62 
      residual.sugar            chlorides  free.sulfur.dioxide 
                0.29                 0.73                 0.75 
total.sulfur.dioxide              density                   pH 
                0.72                 0.30                 0.21 
           sulphates              alcohol 
                0.56                 0.27 

Bartlett's Test p-value: 0
```




```
Parallel analysis suggests that the number of factors =  5  and the
number of components =  NA
```

```
Factor Analysis Results:
Factor Analysis using method =  ml
Call: psych::fa(r = factor_data, nfactors = n_factors, rotate =
"varimax",
    fm = "ml")
Standardized loadings (pattern matrix) based upon correlation matrix
                       ML1   ML2   ML3   h2     u2 com
fixed.acidity         0.65  0.09  0.75 1.00 0.0050 2.0
volatile.acidity      0.60  0.08 -0.24 0.42 0.5804 1.3
citric.acid          -0.13  0.07  0.53 0.31 0.6929 1.1
residual.sugar       -0.36  0.76  0.07 0.71 0.2890 1.4
chlorides             0.48  0.20 -0.04 0.27 0.7282 1.3
free.sulfur.dioxide  -0.64  0.29  0.14 0.51 0.4901 1.5
total.sulfur.dioxide -0.74  0.34  0.17 0.69 0.3064 1.5
density               0.40  0.90  0.16 1.00 0.0049 1.5
pH                    0.25 -0.01 -0.55 0.37 0.6305 1.4
sulphates             0.45  0.08 -0.01 0.21 0.7858 1.1
alcohol              -0.06 -0.74  0.01 0.55 0.4542 1.0

                       ML1  ML2  ML3
SS loadings           2.55 2.19 1.29
Proportion Var        0.23 0.20 0.12
Cumulative Var        0.23 0.43 0.55
Proportion Explained  0.42 0.36 0.21
Cumulative Proportion 0.42 0.79 1.00

Mean item complexity =  1.4
Test of the hypothesis that 3 factors are sufficient.

df null model =  55  with the objective function =  5.71 with Chi
Square =  37094.62
df of  the model are 25  and the objective function was  1.43

The root mean square of the residuals (RMSR) is  0.07
The df corrected root mean square of the residuals is  0.1

The harmonic n.obs is  6497 with the empirical chi square  3393.35
with prob <  0
The total n.obs was  6497  with Likelihood Chi Square =  9289.52  with
prob <  0

Tucker Lewis Index of factoring reliability =  0.45
RMSEA index =  0.239  and the 90 % confidence intervals are  0.235
0.243
BIC =  9070.04
Fit based upon off diagonal values = 0.94
Measures of factor score adequacy
                                                   ML1  ML2  ML3
Correlation of (regression) scores with factors   0.95 0.99 0.96
```

```
Multiple R square of scores with factors          0.91 0.99 0.93
Minimum correlation of possible factor scores     0.81 0.97 0.86

Factor Loadings:

Loadings:
                     ML1    ML2    ML3   
fixed.acidity         0.652         0.750
volatile.acidity      0.598              
citric.acid                         0.535
residual.sugar       -0.360  0.759       
chlorides             0.482              
free.sulfur.dioxide  -0.637              
total.sulfur.dioxide -0.742  0.339       
density               0.402  0.900       
pH                                 -0.554
sulphates             0.455              
alcohol                     -0.736       

                 ML1   ML2   ML3
SS loadings    2.552 2.190 1.290
Proportion Var 0.232 0.199 0.117
Cumulative Var 0.232 0.431 0.548
```



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# SUMMARY AND CONCLUSIONS

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```
cat("\n=== ANALYSIS SUMMARY ===\n")
cat("1. Dataset contains", nrow(wine_data), "observations with",
ncol(wine_data), "variables\n")
cat("2. T-test results: Alcohol content differs significantly between
red and white wines\n")
cat("3. Linear regression R-squared:", round(summary(regression_model)
$r.squared, 3), "\n")
cat("4. Quality classification accuracy:", round(accuracy, 3), "\n")
cat("5. Color prediction test accuracy:", round(test_accuracy, 3), "\
n")
cat("6. Factor analysis extracted", n_factors, "factors explaining
wine properties\n")
```

```
=== ANALYSIS SUMMARY ===
1. Dataset contains 6497 observations with 16 variables
2. T-test results: Alcohol content differs significantly between red
and white wines
3. Linear regression R-squared: 0.361
4. Quality classification accuracy: 0.849
5. Color prediction test accuracy: 0.992
6. Factor analysis extracted 3 factors explaining wine properties
```